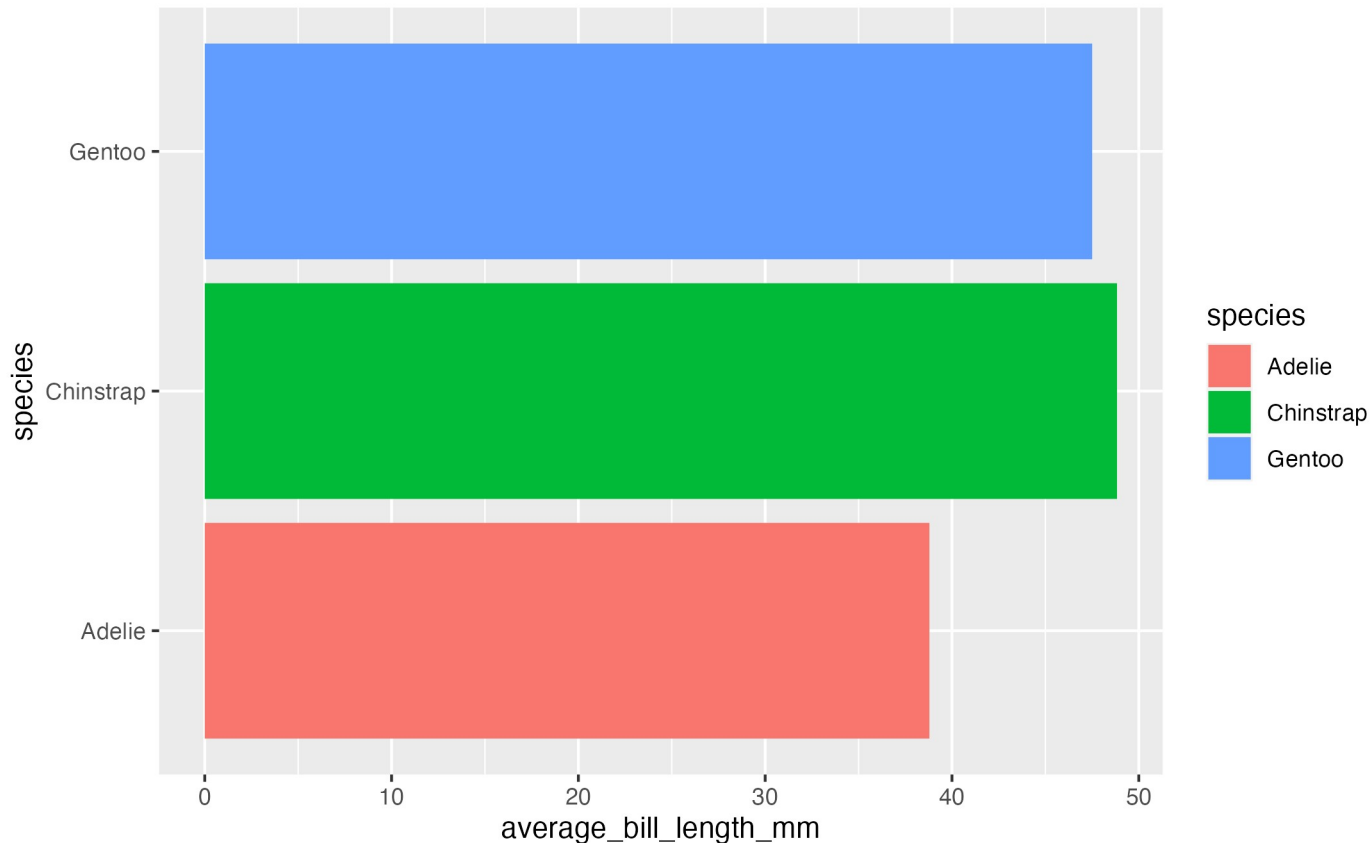


Inferential Statistics

Exercise

The plot below was generated using measurements from a sample of 344 penguins. How do we feel about the following statement:

Based on our data, we can conclude that Chinstrap penguins have longer bills on average than Adelie penguins.



a) **Strongly Agree**

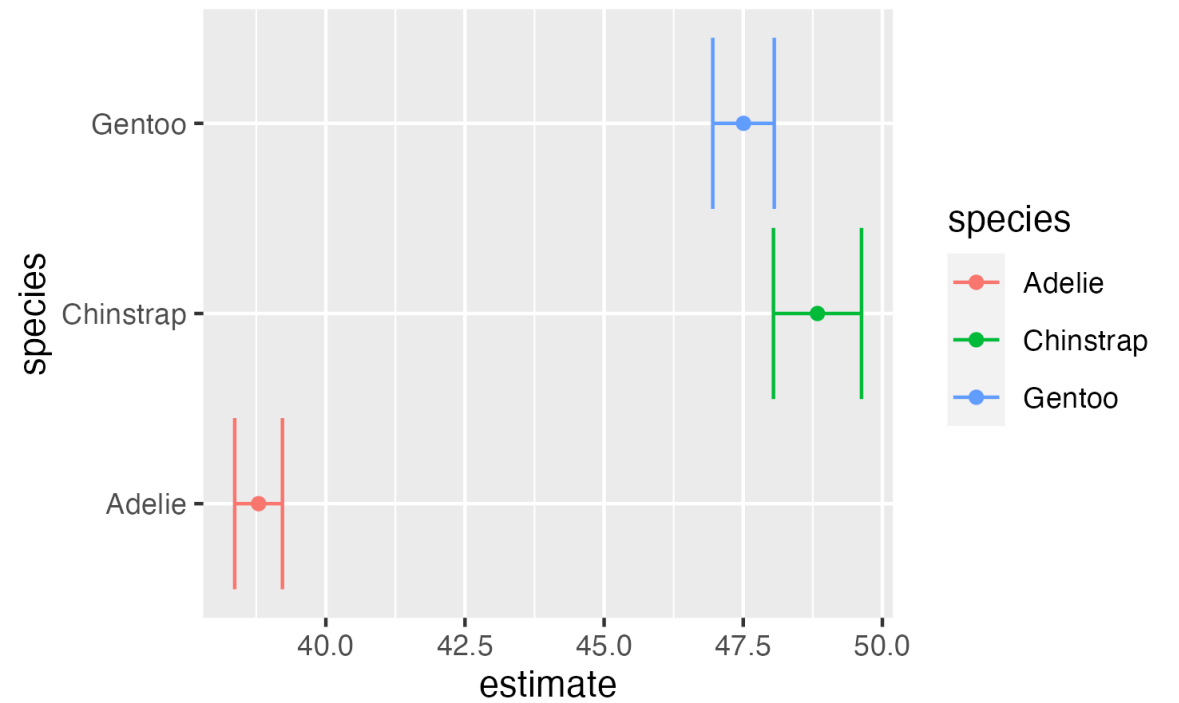
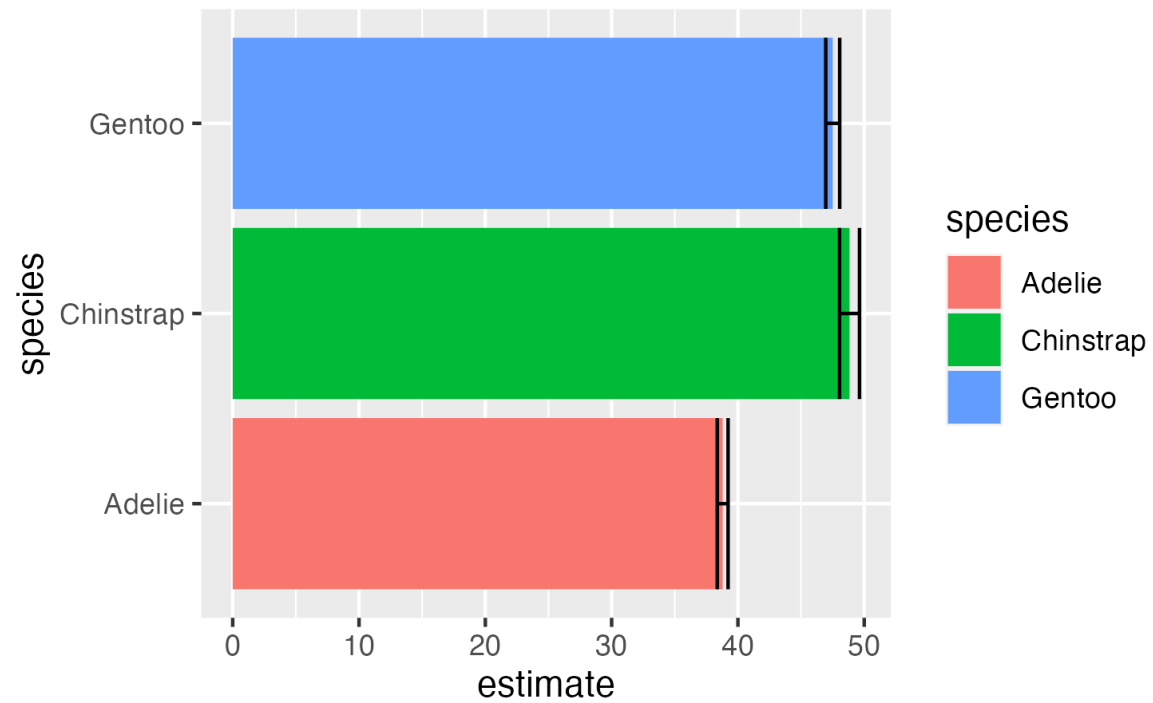
b) **Agree**

c) **Neutral**

d) **Disagree**

e) **Strongly Disagree**

There is some uncertainty around the **true** average bill lengths of Chinstrap penguins and Adelie penguins.



Inferential Statistics

Using sample data to draw conclusions about a population or underlying mechanism, while accounting for uncertainty.

Quantifying what our sample tells us about the population:

- Point Estimate
- Standard Error
- Confidence Interval



University of California
San Francisco